

AfyaSasa — Complete System & Module Audit

****Report date:**** 9 July 2026

****Environment:**** http://localhost:8080 (app) · http://localhost:3000/api/v1 (API)

****Tenant:**** demo (single-hospital mode)

****Automated test result:**** 43 PASS · 3 WARN · 0 FAIL

1. Executive summary

AfyaSasa is a hospital EMR covering reception, outpatient, investigations, inpatient, emergency, specialty services, reporting, and administration. The platform runs on ****Docker**** (PostgreSQL, Redis, MinIO, NestJS API, React frontend, Nginx).

Recent delivery (this cycle):

- Account Center — create/edit users, doctor quick-add, specialisation, password reset, clinical staff directory
- Radiology catalog — CSV template + bulk import (modalities + study protocols)
- Executive Analytics — BI dashboard with date ranges, trends, comparisons, widget config, CSV export
- Auth/session hardening — refresh tokens, cross-tab sync, forced password change without logout
- Single-hospital UX — hospital code hidden on login

****Overall readiness:**** Suitable for ****demo, pilot, and structured user acceptance testing****. Not yet certified for full production at national scale without addressing items in §6 (Improvement roadmap).

2. Platform & infrastructure modules

Module	What it does	Status	Recommended improvements
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PostgreSQL	Primary clinical and admin data store (demo schema)	Working	Full multi-tenant `search_path` on all ORM queries; backup automation
Redis	Session/cache layer	Working	Wire caching for catalog, settings, and report aggregates
MinIO (S3)	Clinical file storage (lab PDFs, radiology attachments, DOCX templates)	Working	Complete `file_registry` integration per enterprise plan
Nginx reverse proxy	Serves frontend on :8080, proxies API	Working	TLS certificates for production; rate limiting
NestJS API	REST API `/api/v1`, Swagger at `/docs`	Working	Global pagination; structured error codes on all list endpoints
React frontend	SPA worklists, forms, control center	Working	Code-split large bundle (>500 KB); deeper mobile QA
Realtime (Socket.IO)	Live settings/worklist updates	Partial	Expand to lab/radiology queue live refresh
Docker Compose	Local and pilot deployment	Working	Production compose + health checks documented

3. Core platform services (backend)

Module	What it does	Status	Recommended improvements
Authentication	Login, refresh rotation, lockout, forced password change	Working	Email password reset in production; MFA enrollment UI
RBAC	Roles, permissions, token revocation on role change	Working	Role templates per job title (nurse, doctor, records)
Tenancy	`X-Tenant` header, pool-level `search_path`	Demo OK	Provisioned tenant schemas for true multi-hospital
Admin / Settings	Hospital profile, clinical catalog, departments	Working	Version history for catalog changes
Audit logs	User action trail, PHI access report	Partial	Before/after JSON snapshots on clinical mutations

Notifications	In-app inbox, SMS config	Working	Delivery status dashboard; push notifications
Clinical orders mirror	Lab + radiology dual-write to unified orders table	Partial	Single order UI for clinicians; pharmacy orders
Worklists API	Cross-department patient queues	Working	Saved filters per user; export
Documents / templates	DOCX render, hospital library	Working	PDF generation parity for all template types
Health check	`/api/v1/health` for ops	Working	Deep checks (DB latency, MinIO, Redis)

4. Clinical workflow modules (main navigation)

4.1 Reception

Screen	Explanation	Status	Improvements
Patient Search	Find existing patients before registration or visit; opens safety banner and profile.	Working	Duplicate detection hints; national ID lookup
Register Patient	Progressive registration with demographics, next of kin, identifiers.	Working	Bulk import from CSV; photo capture
Patient Timeline	Chronological view of encounters, orders, documents, admissions.	Working	PDF export of timeline; filter by module
OPD Check-In	Stepwise check-in: patient → clinic → visit type → confirm encounter.	Working	Queue number printing; appointment auto check-in
Appointments	Schedule, status workflow, calendar views.	Working	SMS reminders; wait-list
Referrals	Internal/external referrals with letter generation and tracking.	Working	eReferral integration; status webhooks
Medical Documents	Patient-linked document repository.	Working	OCR search; category tagging
Sick Sheets	Issue, print, store sick leave certificates.	Working	Digital signature; employer verification QR

4.2 Outpatient (OPD)

Screen	Explanation	Status	Improvements
Triage Queue	Nurse workspace: vitals, triage colour, alerts, prior visits.	Working	NEWS2/MEWS auto-score; device vitals import
OPD Patients	Active outpatients by queue; open chart from worklist.	Working	Panel view by clinic/room
Doctor Queue	Prioritised consultations: SOAP notes, ICD-10 diagnosis, complete visit.	Working	Order sets; e-prescribing module
OPD Reports	Visit summaries and OPD operational reports.	Working	Tie to Executive Analytics drill-down

4.3 Investigations

Screen	Explanation	Status	Improvements
Laboratory	Full lab worklist: request → sample → result → verify; PDF upload.	Working	Analyzer HL7 feed; critical result escalation rules
Lab Patients	Patients with active lab requests (department worklist).	Working	Batch sample collection mode
Lab catalog (admin)	Panels, tests, reference ranges; **CSV bulk import** with template.	Working	LOINC code mapping; price list per test
Radiology	Imaging worklist: request → schedule → report → verify; attachments.	Working	DICOM viewer integration; PACS worklist sync
Imaging Patients	Patients with active radiology requests.	Working	Modality room assignment
Radiology catalog (admin)	Modalities + study protocols; **CSV template import**.	Working	**Import study CSV now** (0 protocols in demo DB)
Results Inbox	Clinician inbox for verified lab/radiology results.	Working	Acknowledge/read receipt; push alert

4.4 Inpatient (IPD)

Screen	Explanation	Status	Improvements
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Inpatient (IPD)	Ward board, admissions, transfers, bed assignment, discharge.	Working	**Fix admissions list API (HTTP 500)** — priority bug
IPD Patients	Current admissions and discharge-ready patients.	Working	Expected discharge date; bed cleaning workflow
ICU	ICU ward view filtered from IPD module.	Working	Ventilator charting; hourly observation grid
HDU	High-dependency unit view.	Working	Step-down criteria checklist
Wards & beds (admin)	Configure wards, bed inventory, occupancy.	Working	Visual bed map designer

4.5 Emergency

Screen	Explanation	Status	Improvements
Emergency	ED command center: triage, bays, observation, disposition.	Working	Real-time ED dashboard TV mode
ED Patients	ED queue by triage and disposition.	Working	Ambulance pre-alert intake
Emergency API	Queue, dashboard, metrics, bays, alerts.	Partial	Bays endpoint requires ED permission (403 for admin — expected)

4.6 Specialty

Screen	Explanation	Status	Improvements
Theatre	Surgical bookings, procedures, scheduling.	Working	WHO checklist; implant tracking
Maternity	ANC, labour, delivery, nursery service line.	Working	Partograph; neonatal integration
Nursing module (API)	Nursing notes and observations (backend).	Partial	Dedicated nursing UI in main nav

4.7 Reports & analytics

Screen	Explanation	Status	Improvements
Clinical Reports	Metric cards, MOH report library (705/706/717), CSV export.	Working	Scheduled email reports

Executive Analytics	BI dashboard: date presets, KPIs, trend charts, period comparison, widget config, CSV export.	Working	Interactive chart library; ward-level drill-down
Operations Center	Today's ops snapshot: OPD, beds, pending labs/radiology, ED.	Working	Auto-refresh intervals configurable
OPD Reports	OPD-specific reporting view.	Working	Merge into unified analytics hub
Worklists	Cross-department operational queues with search.	Working	Role-default landing pages

4.8 Administration

Screen	Explanation	Status	Improvements
Hospital Control Center	Central admin: profile, catalogs, users, security, ops.	Working	Guided setup wizard for new hospitals
Hospital Library	Policies, SOPs, circulars for all staff.	Working	Version control and acknowledgement tracking
Notifications	Staff notification inbox.	Working	Priority levels; mark-all-read

5. Hospital Control Center modules (detailed)

Section	Explanation	Status	Improvements
Hospital profile	Facility name, MOH code, contacts, locale — drives letterhead and defaults.	Working	Multiple sites under one group
Facilities & modules	Enable OPD/IPD/lab/etc. per satellite clinic.	Working	Per-facility go-live checklist
User & access management	Create/edit users, roles, departments, **quick-add doctor** , specialisation, password reset, search.	Working	Bulk user import; SSO/LDAP
Roles & permissions	RBAC bundles, permission matrix.	Working	Permission simulator ("view as role")
Departments & clinics	Clinical departments for OPD routing and reporting.	Working	Link departments to wards and theatres

Clinical catalog	Visit types, payment methods, identifiers, triage defaults.	Working	Import/export catalog JSON
Wards & beds	IPD bed inventory.	Working	Maintenance scheduling
Laboratory catalog	Panels/tests + CSV import template.	Working	Interface to national lab master
Radiology catalog	Modalities + study protocols + CSV import.	Working	Import demo study protocols from template
Theatre configuration	Theatres and procedure catalog.	Working	Block time scheduling rules
Maternity configuration	ANC and delivery templates.	Working	MOH maternity register auto-fill
Document templates	DOCX templates with merge fields.	Working	In-browser template preview
Printing & QR	Patient cards, QR defaults.	Working	National HIE patient ID QR standard
Branding & theme	Logo, colors, favicon.	Working	Dark mode
Notification center	SMS sender, templates.	Working	Africa's Talking / Twilio live credentials
Reports (admin)	Admin entry to reporting tools.	Working	Report permission per director role
Audit logs	Compliance trail.	Partial	Immutable log export to SIEM
Security	Role guidelines reference.	Working	Periodic access review report
Backup & restore	Ops scripts documentation.	Working	Automated nightly backup job
System health	DB, Redis, storage status.	Working	Alerting when service down
Super admin	Platform operator tools.	Working	Restrict to separate super-admin role only

6. Improvement roadmap (prioritised)

Priority 1 — Before pilot go-live

1. ****Fix IPD admissions list**** — `GET /inpatient/admissions` returns HTTP 500; blocks IPD patient worklist reliability.
2. ****Import radiology study protocols**** — use Control Center → Radiology → download CSV template → upload.
3. ****Add clinical staff**** — ensure all doctors have `doctor` role + specialisation in Account Center.
4. ****Mobile/tablet QA**** — verify sidebar, drawer, and full-width content on phones and tablets.
5. ****Production secrets**** — rotate demo passwords; remove `ops/.audit-credentials` from workstations.

Priority 2 — Next development sprint

1. Executive Analytics — ward/clinic drill-down and chart interactivity.
2. Complete `file\_registry` wiring for all uploads.
3. Audit before/after snapshots on clinical writes.
4. Pagination on all list APIs (admissions, encounters, orders).
5. Email password reset and MFA enrollment.

Priority 3 — Enterprise scale (per migration plan)

1. Full multi-tenant ORM (no hardcoded `demo` schema).
2. Unified clinical order engine UI.
3. Redis caching for dashboards and catalogs.
4. Table partitioning for high-volume tenants.
5. National MOH report auto-submission pipelines.

7. Automated test results (9 Jul 2026)

Area	Result
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Infrastructure	5/5 PASS
Authentication	2/2 PASS
Reception	3/3 PASS
Outpatient	3/3 PASS
Investigations	6/6 PASS
Inpatient	2 PASS, 1 WARN (admissions 500)
Emergency	2 PASS, 1 WARN (bays 403 for admin)
Specialty	3/3 PASS
Reports	5/5 PASS
Administration	7/7 PASS
New features	Account center, radiology API, executive analytics — PASS
UI assets	2/2 PASS

8. Manual audit checklist (for director sign-off)

- ☐ Login — hospital code hidden (single-tenant)
- ☐ Reception — register patient → check-in → timeline
- ☐ OPD — triage → doctor queue → complete consultation
- ☐ Lab — order test → enter result → clinician sees Results Inbox
- ☐ Radiology — request imaging → report → verify
- ☐ IPD — admit patient → ward board (verify admissions list loads)
- ☐ ED — triage patient → disposition
- ☐ Account Center — add doctor with specialisation → appears in clinical workflows
- ☐ Radiology catalog — import CSV template
- ☐ Executive Analytics — 30-day range, charts, CSV export
- ☐ Operations Center — today's figures match floor reality
- ☐ Mobile — navigation usable on small screen

9. How to open this report

- **PDF:** ``ops/audit-reports/AfyaSasa-System-Audit-Latest.pdf``
- **HTML (best in browser):** ``ops/audit-reports/AfyaSasa-System-Audit-Latest.html``
- **Chrome:** ``google-chrome ops/audit-reports/AfyaSasa-System-Audit-Latest.html``

AfyaSasa system audit — modules, status, and improvement recommendations.

Regenerated for director review.